Power-factor correction, 3-phase, equivalent-circuit tests

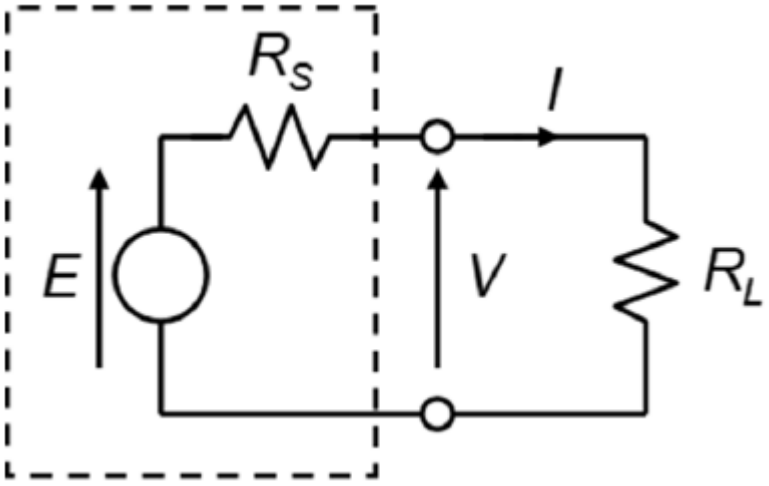
Practice questions · Day 02 · Sun 25 Oct · pack M2

5. Practice

From your course files

Jun23 Q1a (past exam, 5 marks). *Battery Modelling.* The equivalent circuit on the right is used to model a car battery.

i. The battery's open-circuit voltage is measured at 13.8 V. Find the value of the internal voltage source E. [1 mark]
ii. A DC starter motor is connected to the battery. It draws a current of 300 A and the car battery voltage falls to 9.5 V. Find the value of the internal resistance of the battery, R_S, the power dissipated in this resistance, and the power delivered to the motor. [3 marks]
iii. Find the battery output voltage when a load drawing a current of 20 A is connected to it. [1 mark]



Jun23 Q1a figure: battery model E in series with R_S, terminal voltage V, load R_L (from the exam paper)

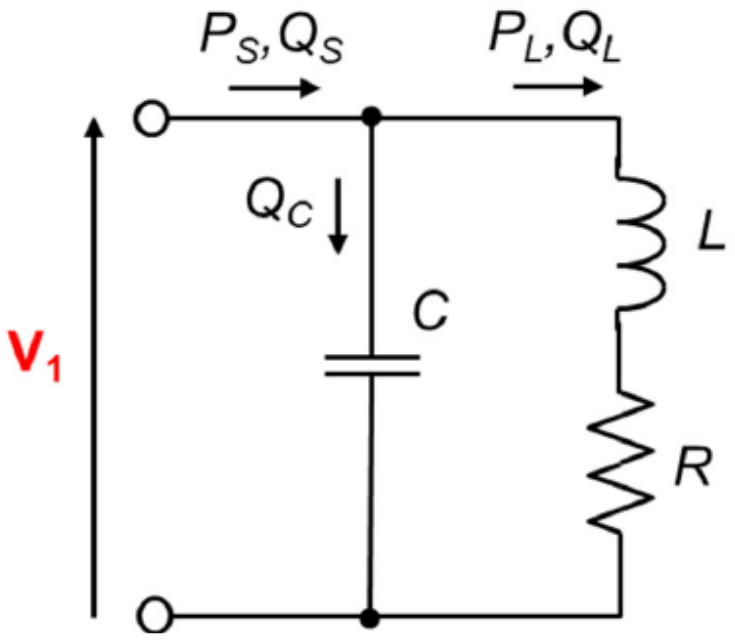
Jun23 Q1b (past exam, 6 marks). *Star-Delta Equivalent Circuits.* Consider a three-phase load consisting of three 70 Ω resistors connected to a 460 V, 3ph supply.

i. If the load is star-connected, calculate the magnitudes of the phase current, line current and total power. [3 marks]

ii. If the load is delta-connected, calculate the magnitudes of the phase current, line current and total power. [3 marks]

Jun24 Q1d (past exam, 8 marks). *Power Factor Correction.* Consider the circuit shown on the right which contains a series RL load which is known to absorb a real power of 80 W and a reactive power of 60 VAr. A power-factor correction capacitor C is connected across the input supply.

i. Sketch the power triangle for the RL load and hence calculate its power factor. [2 marks]
ii. By adding the power factor correction capacitor C, it is desired to have a supply (input) power factor of 0.9 lag at the input to the circuit shown. Find the supply real power P_S and the value of supply reactive power Q_S which gives the desired power-factor. Hence find the value of the reactive power absorbed by the capacitor Q_C. [4 marks]
iii. For another desired value of power-factor, the capacitor is found to absorb −30 VAr. What is the required value of capacitance (in μF) for the above capacitor if the input power supply is 115 V, 60 Hz? [2 marks]

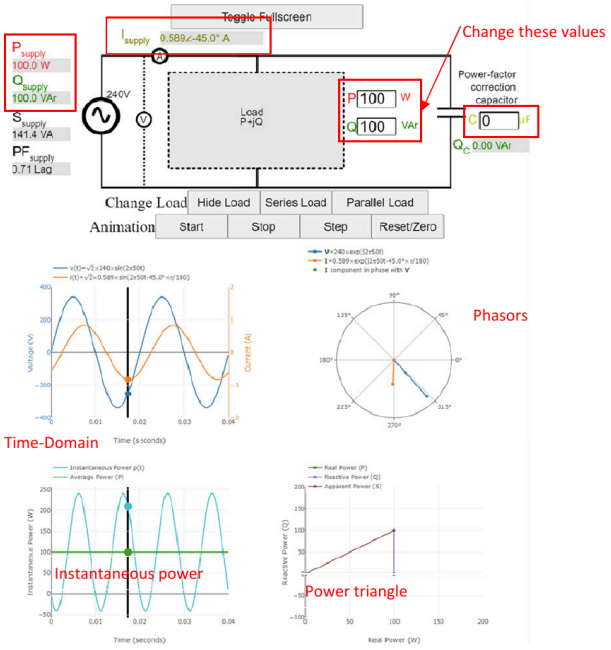


Jun24 Q1d figure: capacitor C across V_1 in parallel with the series L–R load, with P_S, Q_S, Q_C and P_L, Q_L marked (from the exam paper)

Power Circuit Tool WS2 Task 1 (workshop). *Investigating the Load's P and Q Relationship to Supply Current I.* Change the values of P and Q and the record the resultant supply current phasor in the table below.

Case	P (W)	Q (VAr)	$\hat{I}_{supply}$ (A∠°)
1	100	0	
2	0	100	
3	100	100	

1. What is the difference between the supply current phasors for cases 1 and 2?
2. What is the difference between the instantaneous power waveforms between case 1 and 2?
3. Express the supply current phasor for case 3 in terms of the phasors for case 1 and 2? Hint: consider superposition.



Power Circuit Tool WS2 Figure 2: the tool interface, with the P, Q and C inputs and the supply outputs highlighted (from the workshop sheet)

Power Circuit Tool WS2 Task 2 (workshop). *Series and Parallel Equivalent Circuits.* Set the load P = 100 W and Q = 200 VAr. Set "Change Load" to "Series Load" and then "Parallel Load" and record the values in the table below.

Case	$\hat{I}_{supply}$ (A∠°)	R (Ohms)	X (Ohms)
Series			
Parallel			

1. In the series circuit is the voltage or the current the same for both R and X?
2. In the series load, what is the ratio of the values of R and X? Can you explain this using $P = I^2R$ and $Q = I^2X$?
3. In the parallel circuit is the voltage or the current the same for both R and X?
4. In the parallel load, what is the ratio of the values of R and X? Can you explain this using $P = V^2/R$ and $Q = V^2/X$?

Power Circuit Tool WS2 Task 3 (workshop). *Power-Factor Correction.* Set the load P = 100 W and Q = 140 VAr. Set “Change Load” to “Hide Load”. Adjust the magnification of your browser so you can adjust the capacitance and see the supply power triangle at the same time. Record the quantities in the table below with zero capacitance in case 1. Now step the capacitance up in integer steps using the up arrow next to its value. Observe the effect on the current waveform and the power triangle. Find the integer value of capacitance $C_{optimal}$ which results in the smallest supply reactive power and record this as case 3. Then record the values for half this value of capacitance as case 2 and 1.5 times this value of capacitance as case 4.

Case	Capacitance (µF)	P_{supply} (W)	Q_{supply} (VAr)	$\hat{I}_{supply}$ (A∠°)
1: zero capacitance	0			
2: 0.5 $C_{optimal}$				
3: $C_{optimal}$				
4: 1.5×$C_{optimal}$				

1. What is the change in the supply real power as the capacitance is changed? Why is this?
2. Describe the change in the supply current magnitude and angle as the capacitance changes from 0 to $C_{optimal}$ and then beyond $C_{optimal}$.
3. Compare the value of supply current using $C_{optimal}$ versus that for a 240 V source to supply 100 W at unity power-factor.

PQ 2.20 (practice questions). Two single-phase loads $Z_1 = 100 + j0\ \Omega$ and $Z_2 = 10 + j20\ \Omega$ are connected in parallel across a **200 V, 60 Hz** source.

a. Find the magnitude of the two load currents, the total real and reactive power supplied by the source, and the power-factor at the source.
b. Find the capacitance that must be connected in parallel with the loads to improve the power factor to 0.8 lagging.
c. Calculate the total apparent power before and after the capacitor is added.
d. Sketch a power triangle showing apparent, real and reactive power before and after the capacitor is added.

PQ 2.6 (practice questions). A certain car battery can be modelled using an equivalent circuit of a 12.5 V voltage source with a series resistance of 0.02 Ω. Calculate:

a. The open-circuit battery output voltage.
b. The battery current and internal power loss under short-circuit conditions.
c. The battery output voltage and current when powering a load of 1 ohm.
d. When the starter motor is energised, the battery output voltage is observed to fall to 8 V. Calculate the current drawn by the starter motor and the power supplied to the starter motor.
e. Calculate the maximum output power available from the battery.

PQ 2.7 (practice questions). You perform an open-circuit voltage and short-circuit current test on an AA alkaline battery. You measure an open-circuit voltage of 1.61 V and a short-circuit current of 3.1 A.

a. Determine the equivalent circuit parameters E and R.
b. Find its output voltage and power when the load current is 0.5 A.

c. Find the maximum output power from the battery.

PQ 2.21 (practice questions). It is desired to find the equivalent circuit parameters for a single-phase 110 V AC transformer under open-circuit and short-circuit conditions.

a. In the short-circuit condition, the transformer can be modelled by a series resistance and inductive reactance. When 5 V is applied to the transformer, it draws a current of 20 A with a measured input power of 25 W. Find the equivalent parameters for the transformer.
b. In the open-circuit condition, the transformer can be modelled by a parallel resistance and inductive reactance. When 110 V is applied to the transformer, it draws a current of 2.4 A with a measured input power of 40 W. Find the equivalent parameters for the transformer.

PQ 2.26 (practice questions). Each phase of a three-phase load requires a phase voltage of 110 V and a phase current of 10 A. What is the required supply line voltage and current if the load is connected in:

a. star
b. delta

PQ 2.28 (practice questions). A star-connected three-phase resistive load draws 10 A from a 230 V supply.

a. what is the resistance of each phase?
b. The load is reconnected in delta but the supply voltage is unchanged. What are the magnitudes of the new phase voltage, phase current and supply current?
c. By what factors have the phase voltage, phase current and supply current changed?

PQ 2.30 (practice questions). Each phase of a balanced **delta**-connected load consists of a series connected 2 ohm resistor and an inductor with a 3 ohm reactance. It is connected to a 415 V, 3ph supply. Calculate:

a. the phase voltage?
b. the phase current magnitude and phase? (take the phase of the voltage as the reference)
c. the magnitude of the line current?
d. the input power factor?
e. the total input volt-amps (apparent power)?
f. the total input power using the voltage, current and power factor?
g. the total input power as three times the resistive loss $I_1^2 R$ in each phase? (note, this should give you the same result as in the previous part).
h. the total input reactive power?

PQ 2.31 (practice questions). The delta-connected load in the previous question is reconnected in star, however the supply line voltage is unchanged. For each part of the previous question, explain by what factor each quantity changes when connection is changed from delta to star.

Jun23 Q4b (past exam, 7 marks), stretch. *Power Factor Correction.* An industrial plant uses two identical induction motors, each driving a mechanical load of 200 kW. These motors operate with efficiencies of 85% and a power factor of 0.83 lag. One of the motors develops a fault and is replaced with a synchronous motor rated at 250 kVA which has an efficiency of 90%.

i. Calculate the input real and reactive power drawn by the remaining induction motor. [2 marks]

ii. Calculate the input real power and the maximum reactive power capability of the synchronous motor when driving the 200 kW load. [2 marks]
iii. Use the above results to determine the maximum power factor for the plant. [3 marks]

New questions

N1. [4 marks] Three 48 Ω heater elements are connected to a 400 V, three-phase supply.

a. For a delta connection, find the phase current, line current and total power. [2]
b. The elements are reconnected in star. Find the phase voltage, line current and total power, and state the ratio of delta power to star power. [2]

Circuit for New question 2: battery model E and R_S supplying the lights, with the starter motor switched in

N2. [5 marks] A car battery delivers 12.6 V while supplying 5 A to the lights. When the starter motor is engaged the total current is 75 A and the terminal voltage falls to 11.2 V. Use the model E in series with R_S.

a. Find R_S and E. [2]
b. Find the terminal voltage, power to the load and internal loss at 150 A. [2]
c. Find the maximum power the battery could deliver. [1]

Circuit for New question 3: the motor modelled as series R_S and L_S on 230 V, 50 Hz, with the part (d) capacitor across the supply

N3. [6 marks] A single-phase motor on 230 V, 50 Hz draws 6.2 A and 1.1 kW.

a. Find S, Q and the power factor. [2]
b. Find the equivalent series R and L. [2]
c. Find the equivalent parallel R and L. [1]
d. Find the capacitance across the supply needed for unity pf, and the new supply current. [1]

Circuit for New question 4: induction motor, resistive heater and a capacitor bank on the 415 V workshop supply

N4. [8 marks] A workshop on a 415 V, 50 Hz three-phase supply has an induction motor delivering 22 kW of mechanical output at 90% efficiency and 0.82 pf lag, and a 10 kW balanced resistive heater.

a. Find the total P, Q and pf of the workshop, and the line current. [3]
b. Find the reactive power a capacitor bank must supply to raise the pf to 0.98 lag. [2]
c. Find the capacitance per phase if the bank is connected in delta, and if it is connected in star. Which needs the higher voltage rating? [2]
d. The delta-sized capacitors are wired in star by mistake. What pf results? [1]

N5. [7 marks] (Uses your Prac 1 data.) The series-RL prac load absorbs 25.0 W and +31.0 VAr from a 60.5 V, 50 Hz supply.

a. Find the supply current and pf without correction. [1]
b. Find the **two** values of parallel capacitance that give a supply pf of 0.9. State whether each gives a leading or lagging pf. [3]
c. Find the supply current for both values in (b), and the minimum possible supply current. [1]
d. In Table 5 the 36 µF row shows supply Q = +16.6 VAr with load Q = 31.0 VAr and capacitor Q = −45.3 VAr. Explain why the sign must be wrong, and state whether the 0.837 pf is leading or lagging. [2]

6. Quick recall check

1. A 415 V three-phase supply feeds a star load. What is the phase voltage?
2. A balanced load is moved from star to delta on the same supply. By what factor do the line current and the total power change?
3. A load takes 60 W at 0.6 pf lag. What Q_C gives unity pf?
4. A battery reads 12.0 V on open circuit and 11.4 V at 30 A. What is R_S?
5. For the same total Q_C on the same supply, how does the capacitance per phase of a star bank compare with a delta bank?